Group Project

Use R for Statistics
What factors effect E-Commerce Sales

Course: R Language for Statistics (1001)

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Contents

1 Introduction **1**

2 Data Preparation **1**
2.1 Check Missing Data 1
2.2 Transform Data Type 1

3 Visualization **1**
3.1 Time Series 1
3.2 Scatter Plot 3

4 Descriptive Statistics **4**
4.1 Dataset Description 4
4.2 Advanced Analysis 6
4.2.1 Measure of Shape 6
4.2.2 Outlier Detection 6

5 Paired t-Test **7**
5.1 Motivation 7
5.2 Objective Analysis 8
5.3 Process 8
5.4 Result Analysis 9

6 ANOVA Test **9**
6.1 Background and Objective 9
6.2 Modeling 10
6.3 Problem Solving 12

7 Multiple Linear Regression **13**
7.1 Method 13
7.1.1 Interaction Model 13
7.1.2 Least Square Method 14
7.2 Analysis and Evaluation 14
7.2.1 Correlations & Multicollinearity 14
7.2.2 Model Establishment 15
7.3 Model Assumption 18
7.3.1 Linearity 18
7.3.2 Normality 18
7.3.3 Homoscedasticity 19

8 Conclusion **20**

1 Introduction

E-commerce has become a crucial component of modern business, highlighting the need to identify factors influencing sales. This study utilizes R to apply algorithms to analyze a dataset from Tianchi, an e-commerce platform, focusing on the impact of consumption intensity on shopping festivals, the COVID-19 pandemic, and customer behavior on sales performance. To ensure privacy, an anonymized product was randomly selected, disregarding specific product-related details. The analysis employs MEASURES OF LOCATION, PAIRED T-TESTS, ANOVA, and LINEAR REGRESSION to examine these factors and their significance in shaping E-Commerce sales trends.

Keywords: consumption, intensity, COVID-19 pandemic, customer behavior, measures of location, paired t-test, ANOVA, linear regression

2 Data Preparation

2.1 Check Missing Data

Initially, we utilize `is.na()` to check if there are missing data and `sapply()` to obtain all dataset value, the values are all 0 which refers to no missing value. Therefore, we do not need to handle this:

```
              Date         count_view      count_visitor      avg_stay_time
                 0                  0                  0                  0
       bounce_rate        count_buyer         count_deal         money_deal
                 0                  0                  0                  0
count_deal_product
                 0
```

2.2 Transform Data Type

In this dataset, date variable is a `str` type, thus we should transform date from `str` into `datetime` by using `as.Date`.

```
data$Date <- as.Date(data$Date, format = "%Y/%m/%d")
```

3 Visualization

The primary of visualization is not only to explain the data, but also to provide an opportunity for us to conduct subsequent statistical analyses. Therefore, we only visualize some variables which have great influence on our research.

3.1 Time Series

First of all, `money_deal`, representing sales revenue, directly reflects the income level of the online store. Therefore, we visualize this variable utilizing the `geom_line()` function

with the horizontal axis set as `Date` and vertical axis set as `money_deal`. To beautify and clarity the plot, we adjust the parameter `color` and use `labs()` function to add a title.

```
ggplot(data, aes(x = Date, y = money_deal)) +
  geom_line(color = "#4A90E2") +
  theme_minimal() +
  labs(title = "Time Series of Money Deal", x = "Date", y = "Money Deal")
```

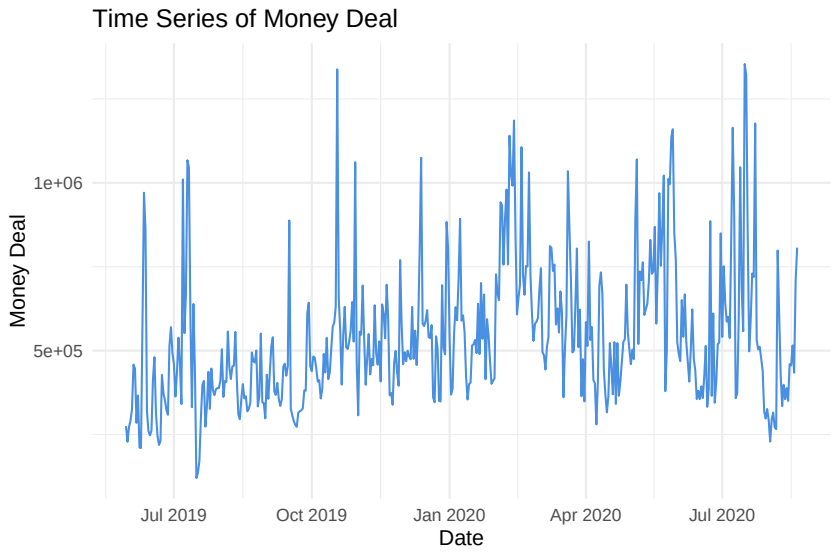


Figure 1: Time Series of `money_deal`

Additionally, `count_view`, representing the views of this online store, indicates user's level of interest in our products. High views generally imply greater potential conversion opportunities Shi, 2023. Therefore, we choose `count_view` to visualization. In a similar manner, we use the `geom_line()` function with the horizontal axis set as 'Date' and vertical axis set as `count_view`. To beautify and clarity the plot, we adjust the parameter `color` and use `labs()` function to add a title.

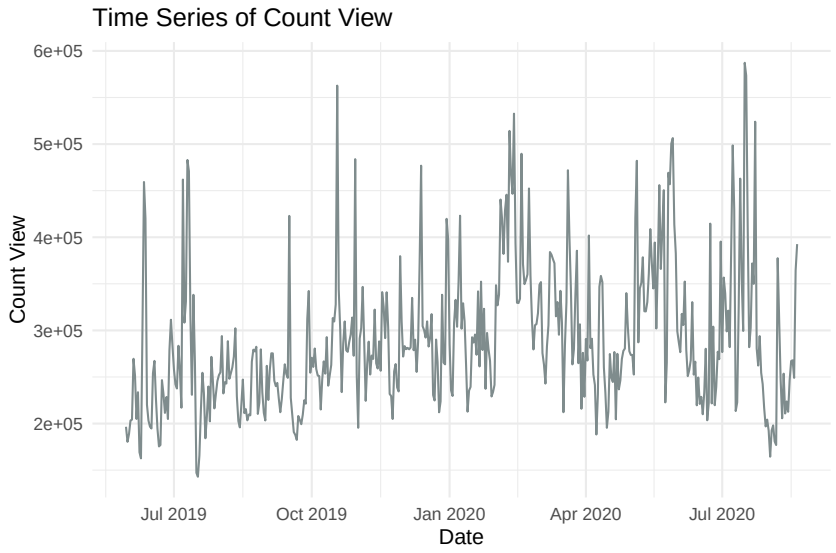
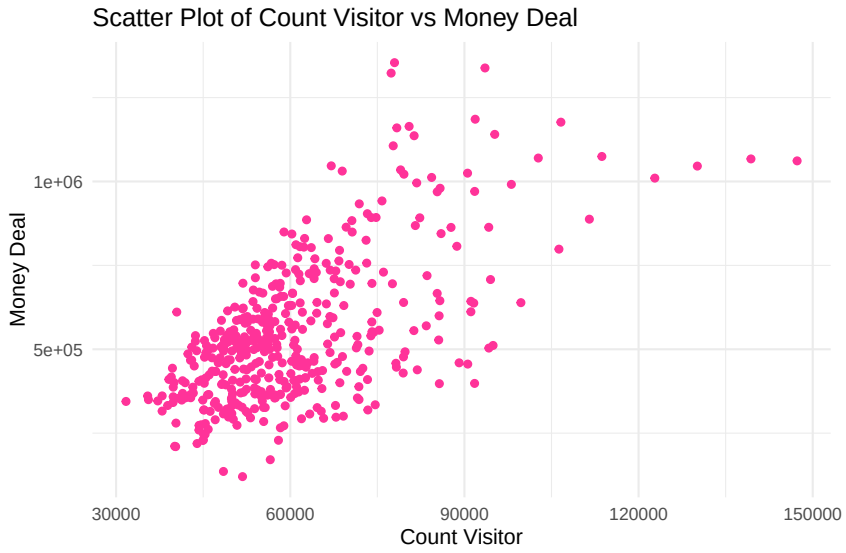
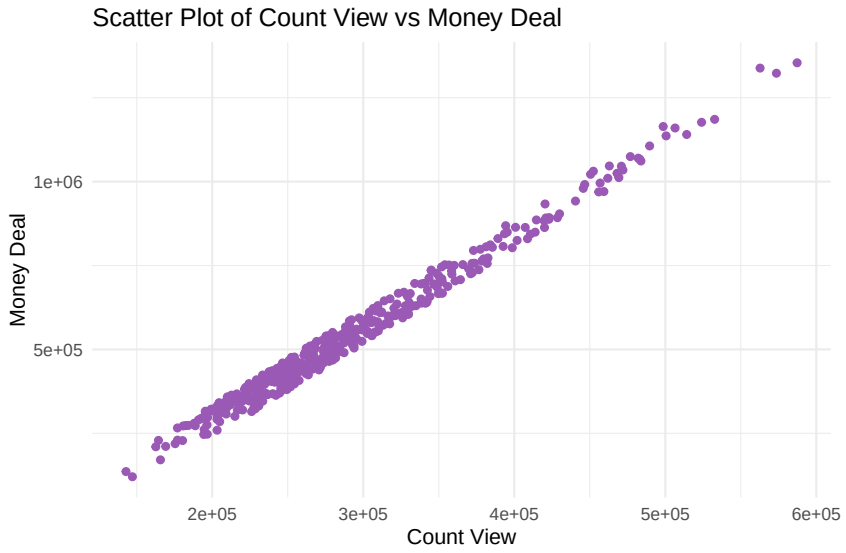


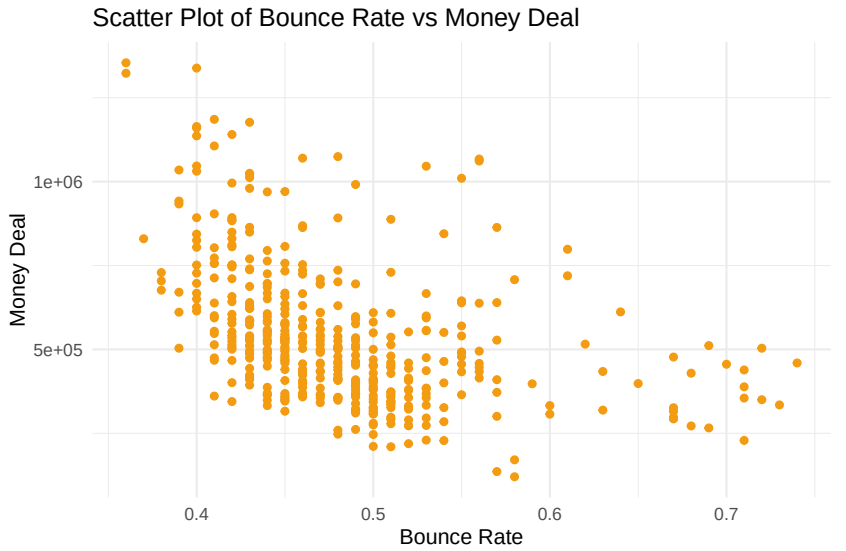
Figure 2: Time Series of `count_view`

3.2 Scatter Plot

To prepare for the subsequent analysis of linear regression, we have created three scatter plots of the variable concerning `money_deal` to observe their relationship preliminarily.







The rest of the histogram and box-plot will be shown later.

4 Descriptive Statistics

In the descriptive statistics section, we intend to utilize various measures to describe dataset and outliers detecting by right of box-plot method.

4.1 Dataset Description

Initially, we obtain data from CSV file:

	Date	count_view	count_visitor	avg_stay_time	bounce_rate	count_buyer	count_deal	money_deal	count_deal_product
	<date>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	2019-05-30	196535	44672	112.	0.5	2994	3159	275033.	10664
2	2019-05-31	180378	44937	112.	0.54	2619	2703	228360.	8906
3	2019-06-01	188768	44374	116.	0.52	2897	2992	272450	10063
4	2019-06-02	203239	50192	110.	0.51	3303	3424	291375.	11551
5	2019-06-03	204189	49664	107.	0.5	3457	3548	325281.	12235
6	2019-06-04	269554	78203	106.	0.56	4357	4532	458280.	16040

Meanwhile, we use ‘`glimpse()`‘ to obtain general dataset information and learn that the all variables (except date) are continuous variables which are suitable to directly analyze. Meanwhile, `avg_stay_time`, and `count_deal_product` are not relevant to our project, so we just delete them:

```
Rows: 449
Columns: 9
$ Date               <date> 2019-05-30, 2019-05-31, 2019-06-01, 2019-06-02, 2019-06-03, 2019-06-04, 2019-06-05, 201...
$ count_view         <dbl> 196535, 180378, 188768, 203239, 204189, 269554, 251798, 204940, 233657, 169181, 162636, ...
$ count_visitor      <dbl> 44672, 44937, 44374, 50192, 49664, 78203, 62722, 55423, 58290, 40098, 40259, 74060, 9176...
$ avg_stay_time      <dbl> 112.02, 112.26, 115.53, 110.10, 107.26, 106.43, 126.01, 104.82, 129.41, 124.43, 121.18, ...
$ bounce_rate        <dbl> 0.50, 0.54, 0.52, 0.51, 0.50, 0.56, 0.51, 0.54, 0.49, 0.50, 0.51, 0.50, 0.45, 0.46, 0.49...
$ count_buyer        <dbl> 2994, 2619, 2897, 3303, 3457, 4357, 3851, 4077, 3669, 2838, 2933, 4227, 6874, 6126, 3447...
$ count_deal         <dbl> 3159, 2703, 2992, 3424, 3548, 4532, 4028, 4216, 3811, 2972, 3073, 4393, 7362, 6600, 3590...
$ money_deal         <dbl> 275033.0, 228359.5, 272450.0, 291374.6, 325280.9, 458279.9, 446472.6, 284660.9, 366155.5...
$ count_deal_product <dbl> 10664, 8906, 10063, 11551, 12235, 16040, 14884, 11736, 13382, 8529, 8161, 17023, 24390, ...
```

And here are explanation for each variable:

- `count_view`: Total daily page views, including repeat visits
- `count_visitor`: Number of unique visitors per day
- `bounce_rate`: Percentage of visits where users leave after one page.
- `count_buyer`: Unique users who made purchases each day.
- `count_deal`: Total number of orders placed daily.
- `money_deal`: Total sales revenue generated each day.
- `count_deal_product`: Total items sold through orders daily.

In order to learn dataset at first, `summary()` is used to showcase data for measure of location and measure of dispersion roughly. The table with summary function is as follows:

```
      Date              count_view     count_visitor    avg_stay_time     bounce_rate      count_buyer
 Min.   :2019-05-30   Min.   :142895   Min.   : 31704   Min.   : 31.19   Min.   :0.3600   Min.   : 1192
 1st Qu.:2019-09-19   1st Qu.:234470   1st Qu.: 49930   1st Qu.: 81.53   1st Qu.:0.4400   1st Qu.: 4419
 Median :2020-01-09   Median :275881   Median : 56734   Median : 88.64   Median :0.4700   Median : 5497
 Mean   :2020-01-09   Mean   :289573   Mean   : 60264   Mean   : 89.90   Mean   :0.4816   Mean   : 6080
 3rd Qu.:2020-04-30   3rd Qu.:327897   3rd Qu.: 66889   3rd Qu.:101.48   3rd Qu.:0.5100   3rd Qu.: 7404
 Max.   :2020-08-20   Max.   :587284   Max.   :147311   Max.   :150.37   Max.   :0.7400   Max.   :17856

   count_deal      money_deal      count_deal_product
 Min.   : 1418   Min.   : 120489   Min.   :  5879
 1st Qu.: 4746   1st Qu.: 388494   1st Qu.: 20328
 Median : 6487   Median : 500704   Median : 31450
 Mean   : 6959   Mean   : 538956   Mean   : 36400
 3rd Qu.: 8519   3rd Qu.: 630138   3rd Qu.: 46410
 Max.   :19944   Max.   :1354130   Max.   :175986
```

With above result, we discover some features:

1. Every mean is larger than corresponding median which indicates that a trend of right skewness with measure of location.
2. The large range computed by maximum value minus minimum value and small IQR calculated by the third percentile minus the first percentile illustrate that there probably are outliers in these variables.

Therefore, we should implement the measure of shape and the outlier detection for deeper analysis.

4.2 Advanced Analysis

`money_deal` is variable we focus on greatly, thus we choose it as the analysis object.

4.2.1 Measure of Shape

In this part, Skewness and Kurtosis are introduced for demonstrate a detailed result which includes skewness and peak-ness or flatness respectively.

For Skewness section, we utilize `skew()` to compute the coefficient of skewness which the formula is:

$$CS = \frac{\frac{1}{N}\sum_{i=1}^{N}(x_i - \mu)^3}{\sigma^3} \tag{1}$$

And obtain result $CS = 1.7$, which is positive and larger than 1. Hence, the shape of `money_deal` has high degree of right skewness. In this way, we can infer that few major shopping events like double eleven indeed influence the overall `money_deal` distribution.

For Kurtosis section, we utilize `kurtosis()` to compute the coefficient of kurtosis which the formula is:

$$CK = \frac{\frac{1}{N}\sum_{i=1}^{N}(x_i - \mu)^4}{\sigma^4} \tag{2}$$

And obtain result $CK = 1.53$ in R but we should add 3, thus the ultimate result $CK = 4.53$, which is larger than 3. Hence, we can conclude that the data is somewhat peaked with less dispersion and may contain more outliers than expected.

And the following histogram can be used to verify:

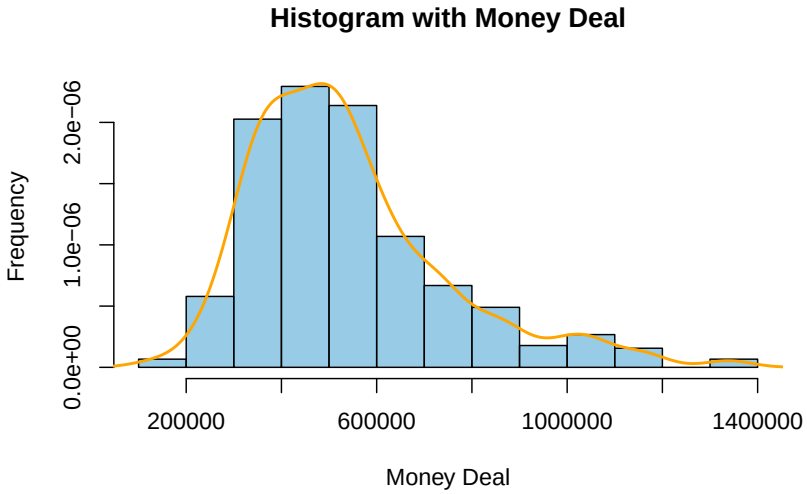


Figure 3: Histogram of `money_deal`

4.2.2 Outlier Detection

In this detection, we initially draw box-plot for `money_deal`:

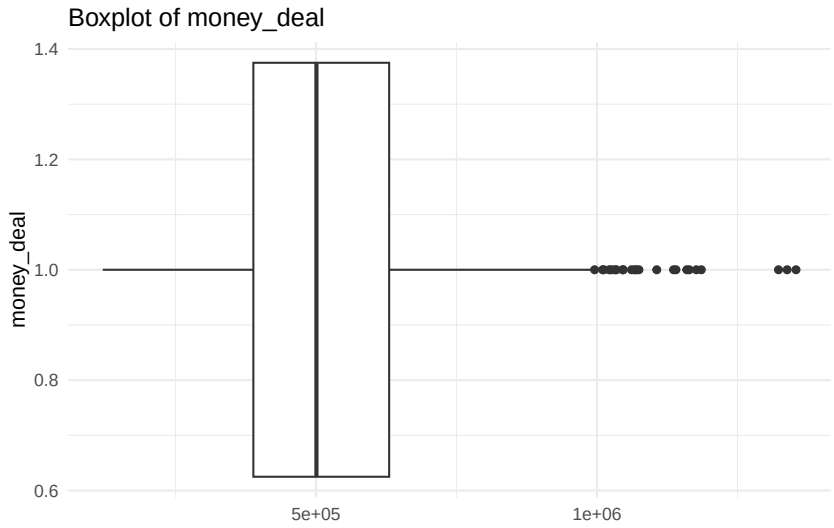


Figure 4: Box-plot of `money_deal`

Then, we handle the outlier issue by IQR method. We calculate $3IQR$ for detecting extreme outliers. With lower and upper fences, we have deleted the overall outliers:

$$x < Q_1 - k\,(Q_3 - Q_1) \quad \text{or} \quad x > Q_1 + k\,(Q_3 - Q_1) \tag{3}$$

where $k = 3$.

Ultimately, we have final dataset, and no outliers are deleted which means they are mild outliers. We do not need to delete mild outliers by right of avoiding data distortion:

	Date	count_view	count_visitor	avg_stay_time	bounce_rate	count_buyer	count_deal	money_deal	count_deal_product
	<date>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	2019-05-30	196535	44672	112.	0.5	2994	3159	275033.	10664
2	2019-05-31	180378	44937	112.	0.54	2619	2703	228360.	8906
3	2019-06-01	188768	44374	116.	0.52	2897	2992	272450	10063
4	2019-06-02	203239	50192	110.	0.51	3303	3424	291375.	11551
5	2019-06-03	204189	49664	107.	0.5	3457	3548	325281.	12235

With previous work done, we can go deeper into our further analysis.

5 Paired t-Test

In this part, we draw a time series curve of `money_deal` through date, discover the trend of sales, extract two peak sales around double-eleven and 6.18 shopping festivals, analyze consumption intensity in these two shopping festivals by paired test, and verify the difference.

5.1 Motivation

In order to observe the trend of product sales, we draw a time series curve of `money_deal` from 2019/5/30 to 2020/8/20.

From the curve, it is apparent that there are two peak values separately near 2019/11/11 and 2020/6/18 which represent 2 popular shopping festivals.

5.2 Objective Analysis

Then, it can be found that the sales during the double-eleven festival are slightly higher than sales during 618 festivals. Then, we assume that sales on and before 2020/6/18 are lower than sales on and before 2019/11/11. Therefore, to verify our assumption, we choose paired t-test to do our experiment.

Paired t-test is an elaboration of two-sample test used to compare the means of the same group at two different times. In this scenario, It aims to compare the means of sales at two different time ranges which fall on double 11 and 618, matching the principle of paired t-test.

5.3 Process

Starting our solutions, first set null hypothesis: the sales on and before 2020/6/18 are equal to or higher than the sales on and before 2019/11/11, over 16 days.

Then, set alternative hypothesis: the sales on and before 2020/6/18 is lower than the sales on and before 2019/11/11.

$$H_0 : x_1 - x_2 \geq 0, \quad H_a : x_1 - x_2 < 0, \quad x_d = x_1 - x_2 \tag{4}$$

And we originally set our significance level $\alpha = 0.05$.

Then, calculate the critical value

$$t = \frac{\bar{d} - x_d}{s_d/\sqrt{n}} \tag{5}$$

where

$$\bar{d} = \frac{\sum d_i}{n}, \quad s_d = \sqrt{\frac{\sum \left(d_i - \bar{d}\right)^2}{n-1}}, \quad n = 16$$

And here we would like to solve the test by using R:

```
# Extract data from 2020/6/3 to 2020/6/18
start_date_1 <- as.Date("2020-06-18") - 15
end_date_1 <- as.Date("2020-06-18")
data_1 <- data[(data$Date >= start_date_1 & data$Date <= end_date_1), ]

# Extract data from 2019/10/27 to 2019/11/11
start_date_2 <- as.Date("2019-11-11")  - 15
end_date_2 <- as.Date("2019-11-11")
data_2 <- data[(data$Date >= start_date_2 & data$Date <= end_date_2), ]

# Make sure the number of rows are the same
if (nrow(data_1) == nrow(data_2)) {
```

```
  # paired test
  paired_test_result <- t.test(data_1$money_deal, data_2$money_deal,
                               paired = TRUE, alternative = "less")
  print(paired_test_result)
} else {
  print("The number of rows are not the same. Paired test cannot be done.")
}
```

5.4 Result Analysis

By achieving by R language code, the output is shown below:

```
Paired t-test

data:  data_1$money_deal and data_2$money_deal
t = -2.199, df = 15, p-value = 0.02199
alternative hypothesis: true mean difference is less than 0
95 percent confidence interval:
      -Inf -13550.61
sample estimates:
mean difference
      -66818.99
```

Comparing critical value t with α, p-value with significance level

$$t = -2.199 < t_{\alpha;\ df=15} = -1.753 \quad \text{or} \quad p\text{-value} = 0.02199 < 0.05 \tag{6}$$

Therefore, H_0 is rejected, H_a can be accepted. It is concluded that the assumption we made: sales on and before 2020/6/18 are lower than sales on and before 2019/11/11 has been verified. Then, it can be reasonably conjectured that the consumption intensity on double eleven is higher than that on 618.

6 ANOVA Test

6.1 Background and Objective

After demonstrating the e-commerce sales exhibits significant eventfulness, we aim to explore whether the outbreak of COVID-19 plays a role Wei et al., 2024. Hence, by utilizing the selecting function, we divide our data into five quarters, three months each, which includes the period before the pandemic, the period of the pandemic and the period during the pandemic.

```
# Date Range 1: From 2019-06-01 to 2019-08-31
start_date_1 <- as.Date("2019-06-01")
end_date_1 <- as.Date("2019-08-31")
data_1 <- data[(data$Date >= start_date_1 & data$Date <= end_date_1), ]

# Date Range 2: From 2019-09-01 to 2019-11-30
start_date_2 <- as.Date("2019-09-01")
end_date_2 <- as.Date("2019-11-30")
data_2 <- data[(data$Date >= start_date_2 & data$Date <= end_date_2), ]

# Date Range 3: From 2019-12-01 to 2020-02-29
start_date_3 <- as.Date("2019-12-01")
end_date_3 <- as.Date("2020-02-29")
data_3 <- data[(data$Date >= start_date_3 & data$Date <= end_date_3), ]

# Date Range 4: From 2020-03-01 to 2020-05-31
start_date_4 <- as.Date("2020-03-01")
end_date_4 <- as.Date("2020-05-31")
data_4 <- data[(data$Date >= start_date_4 & data$Date <= end_date_4), ]

# Date Range 5: From 2020-06-01 to 2020-08-20
start_date_5 <- as.Date("2020-06-01")
end_date_5 <- as.Date("2020-08-20")
data_5 <- data[(data$Date >= start_date_5 & data$Date <= end_date_5), ]

# Combine the data and add a grouping factor
data_combined <- rbind(
  data.frame(data_1, group = "2019-06 to 2019-08"),  # Group 1
  data.frame(data_2, group = "2019-09 to 2019-11"),  # Group 2
  data.frame(data_3, group = "2019-12 to 2020-02"),  # Group 3
  data.frame(data_4, group = "2020-03 to 2020-05"),  # Group 4
  data.frame(data_5, group = "2020-06 to 2020-08")   # Group 5
)
```

6.2 Modeling

Since there are more than three groups, we apply ANOVA to compare the means and identify whether there are any significant differences among them.

The first step of One-Way ANOVA is to construct a hypothesis test with a significance

level $\alpha = 0.05$:

$$H_0 : \mu_1 = \mu_2 = \cdots = \mu_m \quad H_1 : \text{ at least one mean is different from the others} \tag{7}$$

Then, we can use F-statistic to test H_0.

The F-statistical test measures the difference of variation between groups to variance within groups to verify whether it can reject H_0, by first assuming that the data consist of j group, each containing n_i observations, with a total sample size $\sum_{i=1}^{n_j} X_{ij}$. Since the overall mean can be defined as $\bar{\bar{X}}$, we can get the the sample mean of group j is

$$\bar{X}_j = \frac{1}{n_j} \sum_{i=1}^{n_j} X_{ij} \tag{8}$$

and the sample variance of group j is

$$S_j{}^2 = \sum_{i=1}^{n_j} \frac{(X_{ij} - \bar{X}_j)^2}{n_j - 1} \tag{9}$$

and the grand mean is

$$\bar{\bar{X}} = \frac{n_1\bar{X}_1 + n_2\bar{X}_2 + \cdots + n_k\bar{X}_k}{n_1 + n_2 + \cdots + n_k} = \sum_{j=1}^{k} \left(\frac{n_j}{n}\right) \bar{X}_j \tag{10}$$

So, the total sum of squares SST which can be desperate into SSA (Among-Groups Sum of Squares) and SSW (Within-Groups Sum of Squares) given by

$$SST = \sum_{j=1}^{k} \sum_{i=1}^{n_j} (X_{ij} - \bar{\bar{X}})^2 \tag{11}$$

$$SSA = \sum_{j=1}^{k} n_j \left(\bar{X}_j - \bar{\bar{X}}_j\right)^2 \tag{12}$$

$$SSW = (n_1 - 1)S_1{}^2 + (n_2 - 1)S_2{}^2 + \cdots + (n_k - 1)S_k{}^2 \tag{13}$$

To simply the comparisons, the Sum of Squares are converted to Among-Groups Mean Squares and Within-Groups Mean Squares by losing 1 and k degrees of freedom, respectively

$$MSA = \frac{SSA}{k - 1} \tag{14}$$

$$MSW = \frac{SSW}{(n_1 - 1) + (n_2 - 1) + \cdots + (n_k - 1)} = \frac{SSW}{n - k} \tag{15}$$

Ultimately, the F-statistical test is calculated to test the H_0

$$F = \frac{MSA}{MSW} \tag{16}$$

if $F > F_{\text{critical}}$ or p-value $< \alpha$, we can reject H_0 and conclude that it exists some difference among each group.

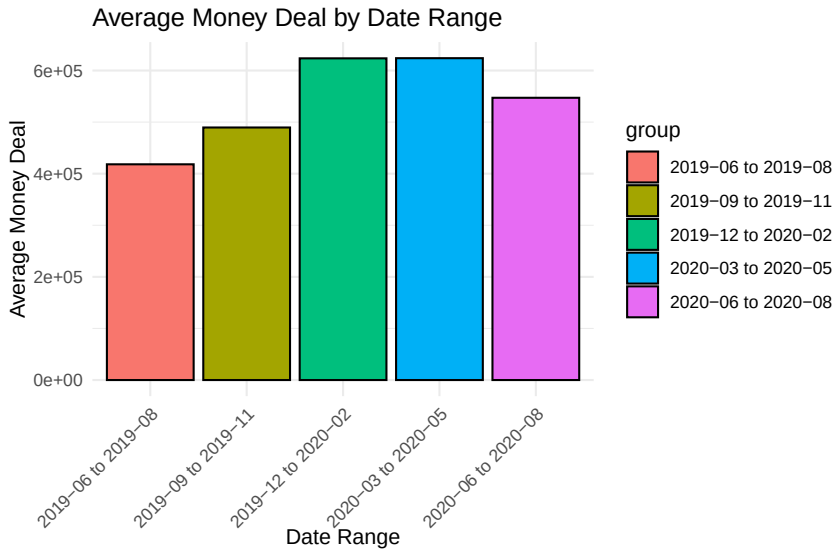


Figure 5: Average `money_deal` by Date Range

6.3 Problem Solving

After realizing the principle of ANOVA, we apply the function `aov()` in R to achieve these mathematical processes.

```
# Perform ANOVA on 'money_deal' with 'group' as independent variable
anova_result <- aov(money_deal ~ group, data = data_combined)

# Output the summary of the ANOVA test result
summary(anova_result)
```

And the results are as followed:

```
             Df    Sum Sq   Mean Sq F value   Pr(>F)
group         4 2.885e+12 7.212e+11   19.22 1.37e-14 ***
Residuals   442 1.659e+13 3.752e+10
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Based on the result, the F value is large so that we can conclude that the variation caused by the independent variable is real and not due to chance. And the $P_r(> F)$ column, represented the p-value of the F-statistic, is much smaller than significance level $\alpha = 0.05$. Therefore, we can reject H_0 and conclude that at least one quarter's mean is different from others.

To further study whether the outbreak have influenced the `money_deal`, we create a bar plot to display the average money deal for each quarter. The plot shows an upward trend in the early period, with a sharp rise in December 2019, but it exists a slight decline

in June 2020. Therefore, we suppose that this pattern can be attributed to the effects of the COVID-19 pandemic. Because in the early stage of the pandemic, hoarding behavior might lead to a rapid rise in `money_deal`, but as the pandemic progressed, the economic recession might lead to a reduction in people's consumption, thereby there is a decline in `money_deal`. So, we suppose COVID-19 is one of the factors affecting e-commerce sales.

7 Multiple Linear Regression

Multiple Linear Regression is a model that estimate the linear relationship between scalar responses (dependent variables) and input vectors (independent variables containing several features)

$$\mathbf{y} = \boldsymbol{\beta}_0 + \boldsymbol{\beta}_1 \mathbf{X} \tag{17}$$

and its expansion formulation could be written as

$$y_i = \beta_0 + \beta_1 X_{i1} + \cdots + \beta_k X_{ik} + \varepsilon_i, \quad i = 1, 2, \ldots, j \tag{18}$$

where j is the number of observations, k is the number of features (inputs), ε_i is the random error term which follows normal distribution (assumption: $E\{\varepsilon_i\} = 0$, $\sigma^2\{\varepsilon_i\} = \sigma^2$ and $\mathrm{Cov}(\varepsilon_i, \varepsilon_j) = 0$ for $\forall\, i, j; i \neq j$).

7.1 Method

Although it is impossible for us to find the mathematical solution that precisely describe the data, but we can find an approximate one with out estimation

$$\hat{y}_i = b_0 + b_1 X_{i1} + \cdots + b_k X_{ik} + e_i, \quad i = 1, 2, \ldots, j \tag{19}$$

where $\hat{y}_i$, b_i and e_i are the point estimators of y_i, β_i, ε_i, respectively.

7.1.1 Interaction Model

However, because the features (inputs) may have interaction, we then should use estimator β to illustrate the interaction between variables. Here we have three features, so the interaction model could be

$$y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \beta_4 X_1 X_2 + \beta_5 X_1 X_3 + \beta_6 X_2 X_3 + \beta_7 X_1 X_2 X_3 \tag{20}$$

Since there are $\sum_{i=0}^{k} \mathrm{C}_n^i$ combinations of features (include no-feature), so there are $\sum_{i=0}^{k} \mathrm{C}_k^i$ parameters β_i.

7.1.2 Least Square Method

The Least Squares is a parameter estimation method of regression evaluation by minimizing the Sum of Squares Errors (SSE)

$$SSE = \sum_{i=1}^{j} \varepsilon_i^2 = \sum_{i=1}^{j} (y_i - \hat{y}_i)^2 \tag{21}$$

Since the distribution of ε_i is assumed to be normal, the sample variance of residuals is an unbiased estimator of the variance σ^2 of an infinite population

$$\sigma^2 \approx s^2 = \frac{SSE}{n - \sum_{i=0}^{k} \mathrm{C}_n^i} = \frac{\sum_{i=1}^{j} (y_i - \bar{y}_i)^2 - 1}{n - \sum_{i=0}^{k} \mathrm{C}_n^i} \tag{22}$$

where the degree of freedom of sample variance is $n - \sum_{i=0}^{k} \mathrm{C}_n^i$ since $\sum_{i=0}^{k} \mathrm{C}_n^i$ degree of freedoms parameters β_i had to be estimated in obtaining the estimated means $\bar{y}$.

This regression model could have small errors (residuals) after taken the least square result while a large difference between an observed value and fitted value is not what we want.

7.2 Analysis and Evaluation

We decided to first select the important variables, then build a regression model, and find a feasible solution by the least squares method.

7.2.1 Correlations & Multicollinearity

Before we calculate the parameter coefficient, we should determine the input features first. The result that we can seek correlation of each input and output is the Correlation Matrix, a table which we can get the relationships and the coefficients:

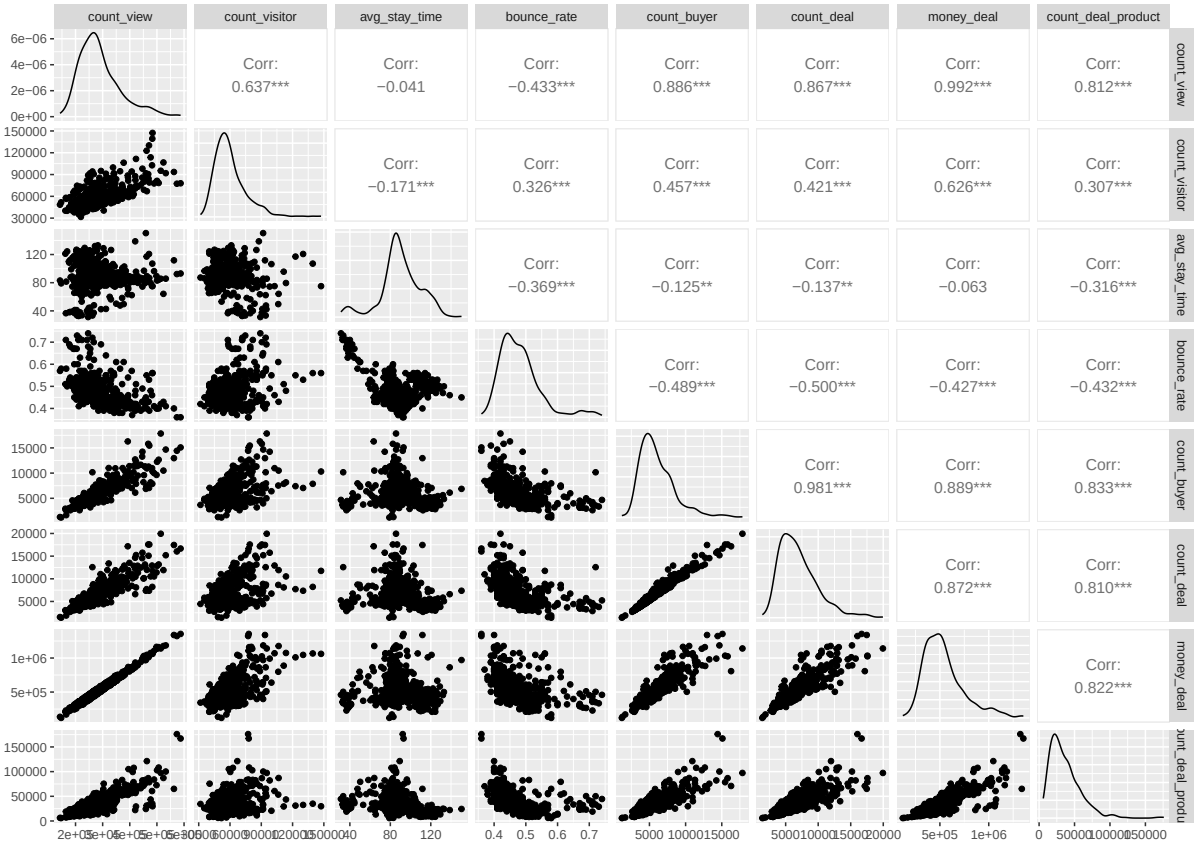


Figure 6: Relationship and Correlation

As we can see, the feature `count_view` has the most absolute correlation

$$\text{Cor}\,(\text{money_deal}, \text{count_view}) = 0.992$$

and the feature `avg_stay_time` has the least absolute correlation

$$\text{Cor}\,(\text{money_deal}, \text{avg_stay_time}) = -0.063$$

so we should use this parameter in priority and exclude `avg_stay_time` first, and then we should seek another parameters that have high correlation with output but low correlation with `count_view` because we should also prevent the multicollinearity which will give non-precise and unstable output. Here we exclude all the features having higher than 0.8 correlation with `count_view`, i.e., we exclude `count_buyer`, `count_deal`, `count_deal_product`, entering `count_visitor`, `bounce_rate` these two more variables.

7.2.2 Model Establishment

Up to now, we have three inputs `count_view`, `count_visitor` and `bounce_rate` with one output `money_deal`. We could establish the interaction regression model like this

based on the assumption that all the features are useful:

$$\begin{aligned} y_i = \beta_0 + \beta_1 \text{count_view} + \beta_2 \text{bounce_rate} + \beta_3 \text{count_visitor} \\ + \beta_4 \text{count_view} \times \text{bounce_rate} \\ + \beta_5 \text{count_view} \times \text{count_visitor} \\ + \beta_6 \text{bounce_rate} \times \text{count_visitor} \\ + \beta_7 \text{count_view} \times \text{bounce_rate} \times \text{count_visitor} \end{aligned} \quad (23)$$

And we use the R to solve the result:

```
Call:
lm(formula = money_deal ~ count_view * bounce_rate * count_visitor,
    data = df)

Residuals:
   Min     1Q Median     3Q    Max
-52002 -21190   1355  18657  57972

Coefficients:
                                    Estimate Std. Error t value Pr(>|t|)
(Intercept)                        -3.050e+04  1.413e+05  -0.216  0.82926
count_view                          2.255e+00  4.729e-01   4.768 2.53e-06 ***
bounce_rate                        -2.546e+05  2.829e+05  -0.900  0.36859
count_visitor                      -6.042e+00  1.977e+00  -3.056  0.00238 **
count_view:bounce_rate              6.882e-01  9.601e-01   0.717  0.47388
count_view:count_visitor            1.213e-05  6.122e-06   1.981  0.04820 *
bounce_rate:count_visitor           7.940e+00  3.917e+00   2.027  0.04328 *
count_view:bounce_rate:count_visitor -1.665e-05  1.214e-05  -1.372  0.17090
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 24840 on 441 degrees of freedom
Multiple R-squared:  0.9861, Adjusted R-squared:  0.9859
F-statistic:  4483 on 7 and 441 DF,  p-value: < 2.2e-16
```

As we can see, we also conduct a F-test for overall significance, which the hypothesis are

$$H_0 : \beta_0 = \beta_1 = \cdots = \beta_k = 0 \quad H_a : \text{At least one not zero} \quad (24)$$

Since the p-value is much smaller than the significant level $\alpha = 0.05$, we have sufficient evidence to reject H_0 and conduct H_a, proving this model is effective.

And we then focus on the t-test for individual significance with hypothesis

$$H_0 : \beta_i = 0 \quad H_a : \beta_i \neq 0 \quad (25)$$

following t distribution with $j - k - 1$ degrees of freedom. And the fact is that we do not have sufficient evidence to prove that $\beta_0, \beta_2, \beta_4, \beta_7 \neq 0$ since their t-value $t_0, t_2, t_4, t_7 >$

0.05, i.e., we cannot consider them as our parameters except intercept because intercept would change to significant. So our updated model is that

$$\begin{aligned} y_i = \beta_0 + \beta_1 \text{count_view} + \beta_2 \text{count_visitor} \\ + \beta_3 \text{count_view} \times \text{count_visitor} \\ + \beta_4 \text{bounce_rate} \times \text{count_visitor} \end{aligned} \quad (26)$$

and the result is

```
Call:
lm(formula = money_deal ~ count_view + count_visitor +
    count_view:count_visitor +
    bounce_rate:count_visitor, data = df)

Residuals:
   Min     1Q Median     3Q    Max
-51431 -21077   1708  18769  58739

Coefficients:
                            Estimate Std. Error t value Pr(>|t|)
(Intercept)               -1.636e+05  1.810e+04  -9.037  < 2e-16 ***
count_view                 2.660e+00  5.927e-02  44.888  < 2e-16 ***
count_visitor             -3.575e+00  5.964e-01  -5.994 4.25e-09 ***
count_view:count_visitor   3.677e-06  8.070e-07   4.556 6.76e-06 ***
count_visitor:bounce_rate  2.745e+00  5.586e-01   4.914 1.26e-06 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 24940 on 444 degrees of freedom
Multiple R-squared:  0.9859, Adjusted R-squared:  0.9858
F-statistic:  7785 on 4 and 444 DF,  p-value: < 2.2e-16
```

And this new model satisfies both F-test for overall significance and t-test for individual significance.

This new model now shows high significance of these parameters, high importance and high correlation with adjusted $R^2 = 0.9858$, indicating that there are 98.58% of output `money_deal` could be explained by the input vector of features, which is a very powerful

regression and prediction:

$$y_i = -1.636 \times 10^5 + 2.66 \text{ count_view} - 3.575 \text{ count_visitors} + 3.677 \times 10^{-6} \text{ count_view} \times \text{count_visitor} + 2.745 \text{ bounce_rate} \times \text{count_visitor} \quad (27)$$

7.3 Model Assumption

After we do linear regression construction and conclude a suitable model, here we should verity the assumption we use the linear regression.

7.3.1 Linearity

First we should verify the assumption that there is the linear relationship between inputs and outputs, that is, the residual (error) of linear model prediction from the observational output is random distributed and follows the line $y = 0$. Here we would like to use the scatter plot with residual and fitted values to verify the linearity

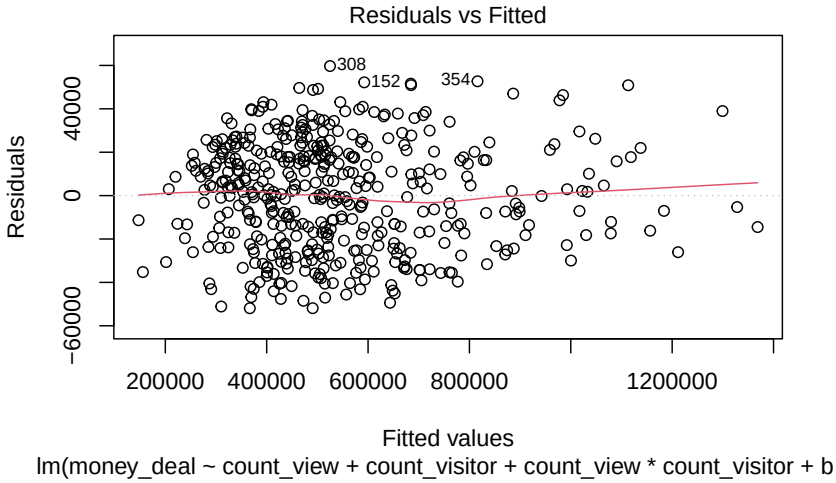


Figure 7: Scatter Plot with Residual & Fitted Values

As we can see, the residuals and the predicted values appear random and appear linear with horizontal line. So, it follows the linearity.

7.3.2 Normality

Although we cannot use the Shapiro-Wilk Test to conclude the normality (according to the central limit theorem, larger sample always decrease the p-value, even though the samples from a very normal likely distribution), this assumption for ε_i is justifiable in this situations because this estimation and testing procedures are based on the t distribution

and are only sensitive to large difference from normality. And if we still want to avoid this problem, graphical methods are recommended. Here we draw the Q-Q plot with residuals and theoretical quantities:

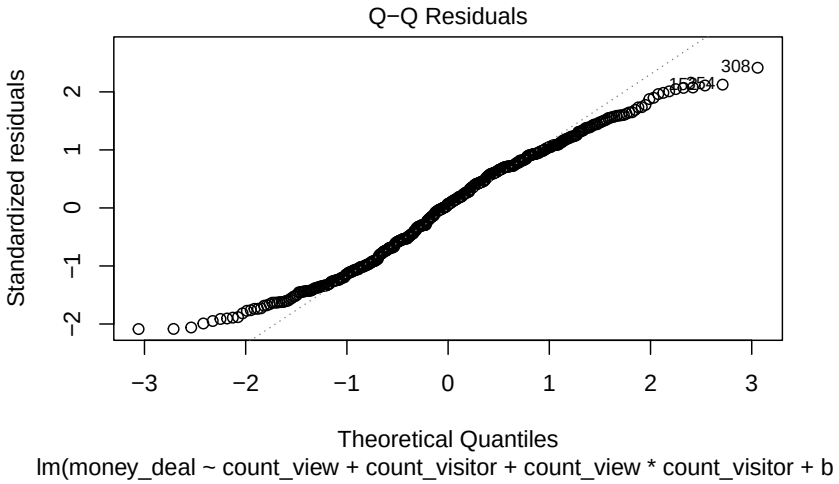


Figure 8: Q-Q Plot with Residual & Quantities

We can see that most of the observations are fitted along the line, so that we can verify the normality of residuals (error).

7.3.3 Homoscedasticity

Since we want to get a model that performs well on average, so we should let the variability of points around the least squares line be roughly constant. We will check this by using the Non-constant Variance Score Test, a χ^2-test with hypothesis

$$H_0 : \text{Homoscedasticity} \quad H_a : \text{Heteroskedasticity} \tag{28}$$

and our result is

```
Non-constant Variance Score Test
Variance formula: ~ fitted.values
Chisquare = 0.3344521, Df = 1, p = 0.56305
```

where the p-value is larger than significant level, therefore, we do not have sufficient evidence to reject the χ^2-test, we could conclude that variability of points around the least squares line should be roughly constant.

After checking the assumption, we can conclude that this model is valid and proven effective.

8 Conclusion

Our findings do provide solutions to improve E-Commerce sales and business strategies.

1. Identified Key Factors: Our research identifies the main factors that influence E-Commerce sales, mainly sales discounts, COVID-19 impacts, and customer behavior.

2. Understanding Economic Trends: Provides insights into how external factors like the pandemic affect consumer spending patterns and sales performance.

3. Improve Sales Strategies: Businesses can design more effective marketing strategies.

The real-world application for our research results are these:

1. E-Commerce Platforms: Platforms of online marketplaces can use our findings to optimize website performance.

2. Business Decision-Making: Retailers can adapt to economic disruptions like pandemics by forecasting sales.

3. Marketing and Customer Engagement: Insights into customer behavior can help improve user experience and retain customers.

4. Policy and Economic Planning: Governments and organizations can use the results for planning support programs or promoting E-Commerce.

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